Cloud-Native Infrastructure & Microservices Architecture Specification

High-Availability Vector Search, Distributed Ingestion & Kubernetes Topology

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1. System Architecture Overview

This specification establishes the architectural blueprints for the ContentAI distributed deployment. The application is structured into decoupled frontend, backend, vector storage, and relational persistence layers designed for sub-second query latency and 99.99% operational uptime.

Layer	Technology Stack	Key Functionality
Frontend UI	Next.js 16 (React 19), Tailwind CSS	Warm editorial design system, interactive Tippy markdown editor, SVG analytics
API Gateway	FastAPI, Uvicorn, ASGI AsyncIO	REST endpoints, JWT authentication, rate limiting (200 req/min), CORS policy
Vector Store	FAISS (Facebook AI Similarity Search)	In-memory L2 index with serialized disk persistence and dynamic reloads
Relational DB	SQLite / PostgreSQL with SQLAlchemy	ACID transaction management, document metadata, audit logging
NLI Verifier	Cross-Encoder Claim Extraction Engine	Dual-phase factual consistency verification and hallucination auditing

2. Reliability, Failover & Graceful Degradation

A foundational design principle of ContentAI is resilient self-healing. When external third-party cloud services experience throttling, authentication failures, or network interruptions, the platform activates local fallbacks:

- **Embedding Fallback:** Automatically degrades to LocalDeterministicEmbeddingService with 1536-dimensional L2-normalized hash representations.
- **LLM Fallback:** Activates MockGroundedLLMProvider, synthesizing grounded text extracted directly from retrieved document chunk context.
- **Zero-Downtime Session Retention:** Client-side localStorage tokens ensure uninterrupted workspace access during transient network reconnects.

3. Security & Compliance Controls

All data at rest is isolated by user identifier. Multi-tenant queries include strict tenant isolation predicates in every SQLAlchemy query filter. File uploads are validated by magic byte detection to eliminate malicious executable injection.